Juan Nathaniel

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Research Experience

NYU, Courant Institute of Mathematical Science — *New York*
Incoming Postdoctoral Researcher — *Anticipated start: 2027*
- Advisor: Laure Zanna

Professional Experience

2023, 2026 **Research Intern**, IBM TJ Watson Research Center, New York
2019 - 2021 **Research Engineer**, Living Analytics Research Center, Singapore

Education

Columbia University — *New York*
PhD, Earth and Environmental Engineering — *Expected 2027*
MS, Earth and Environmental Engineering — *2021–2022*
- Advisor: Pierre Gentine

National University of Singapore — *Singapore*
BS (Hons.), Environmental Science — *2015–2019*

Selected Publications

(*) denotes co-first authorship. Full list can be found here: Google Scholar↗

Published

[1] Sevinchan*, **Nathaniel***, Ueltzhöffer*, et al. (2026). *In-context learning to predict critical transitions in dynamical systems*
Advances in Neural Information Processing Systems 39

[2] **Nathaniel** & Gentine (2026). *Generative emulation of chaotic dynamics with coherent prior*
Computer Methods in Applied Mechanics and Engineering

[3] **Nathaniel**, et al (2025). *Deep Koopman operators for causal discovery*
Communications Physics

[4] Herdeanu* & **Nathaniel***, et al (2025). *CausalDynamics: A large-scale benchmark for discovery of dynamical causal models*
Advances in Neural Information Processing Systems 38

[5] **Nathaniel**, et al (2024). *ChaosBench: A multi-channel, physics-based benchmark for S2S climate prediction*
Advances in Neural Information Processing Systems 37

[6] Qu*, **Nathaniel***, et al (2024). *Deep generative data assimilation in multimodal setting*
Computer Vision and Pattern Recognition (Workshop on EarthVision)

[7] Kim*, **Nathaniel***, et al (2024). *Spatiotemporal upscaling of sparse air-sea pCO2 data via physics-informed transfer learning*
Scientific Data

[8] **Nathaniel**, et al (2023). *MetaFlux: Meta-learning global carbon fluxes from sparse spatiotemporal observations*
Scientific Data

In Prep / In Review

[9] Gentine, Balwada, ⋯, **Nathaniel**, Watson-Parris, and Eyring (2026). *legoESM: a modular, differentiable, multiscale, AI-ready Earth system model built with AI agents*.
arXiv preprint arXiv:2610.11883

[10] **Nathaniel**, et al (2026). *Koopman early warning signals for bifurcation and rate-induced tipping*. *In revision for Communications Physics*

Awards, Fellowships, & Grants

2026	**Center for Nonlinear Studies Postdoctoral Fellowship**, Los Alamos National Lab	*(declined)*
2026	**Best Paper Award**, ICML Workshop on AI4Physics	
2026	**Google Research Award**, Google Research	*$ 100,000*
2025	**Columbia-Dream Sports AI PhD Fellowship**, Columbia University	*$ 100,000*
2024	**Oral Presentation (top 0.6% of 14k+ submissions)**, NeurIPS	
2024	**Best Student Paper Award**, CVPR Workshop on EarthVision	
2019	**Dean's List**, National University of Singapore	
2015 – 2019	**ASEAN Undergraduate Scholarship**, National University of Singapore	*$ 200,000*

Invited Talks

2026 **Koopman Early Warning Signals**, Potsdam Institute for Climate Impact Research
2026 **Emulating, Understanding, and Probing Chaotic Systems**, CNLS, Los Alamos
2026 **Operator Learning for Climate Tipping Point Detection**, UK ARIA
2025 **Deep Learning Koopman Causality**, Potsdam Institute for Climate Impact Research
2025 **Operator Learning for Climate Modeling Across Scales**, University of Utah
2025 **Subseasonal-to-Seasonal Climate Predictability**, Stanford University
2024 **Climate Predictability Across Timescales**, UCLA
2023 **Understanding and Modelling the Earth System with Machine Learning**, DLR
2023 **Meta-learning for Environmental Sciences**, University of Reading

Teaching Experience

2025 **A Better Planet by Design**, Teaching Assistant
2025 **Hackathon: Learning the Earth with Artificial Intelligence and Physics**, Organizer
2024 **Machine Learning for Environmental Engineering and Science**, Teaching Assistant
2024 **Bootcamp: Learning the Earth with Artificial Intelligence and Physics**, Instructor

Service

Reviewer (Climate Science): Journal of Climate, Communications Earth & Environment, Journal of Advances in Modeling Earth Systems (JAMES).

Reviewer (Machine Learning): International Conference on Learning Representations (ICLR), Advances in Neural Information Processing Systems (NeurIPS), International Conference on Machine Learning (ICML).

Reviewer (Interdisciplinary): Nature Communications, Communications Earth & Environment, International Journal of Bifurcation and Chaos (IJBC).